

An Agent-to-Agent Freight Marketplace for Empty Return Trips

with Trucker Coalitions and Manipulation-Resistant Negotiation

In one line: every trucker and every shipper gets an AI agent that negotiates on their behalf; game-theoretic mechanisms keep the negotiation fair and honest; a route optimiser guarantees every deal is physically feasible; and the human gives the final yes or no, like accepting a ride on Rapido, except the price has already been bargained for them.

Domain	Road freight and logistics (India)
SDG targets	9.1 resilient infrastructure · 9.3 access of small enterprises to value chains · 9.4 resource-efficient, cleaner industry
Core techniques	Multi-agent LLM systems · auction and bargaining theory · cooperative game theory (Shapley value) · vehicle routing optimisation · agent security
Deliverable	Working simulation of a city freight market with live dashboard, before/after comparison, and robustness experiments

1. Abstract

Indian road freight is dominated by small operators who own two or three trucks and find return loads through phone calls, WhatsApp groups and commission brokers. Industry estimates suggest roughly a third of trucks return empty after a delivery, and long-haul drivers who refuse to run empty instead wait days for a load. Digital load boards already exist, and AI voice agents already negotiate freight rates, but only on behalf of brokers: the trucker never has an agent of his own. Research on two-sided agent markets shows that when LLM agents trade with each other freely, they fall for manipulation, accept the first offer, and degrade at scale.

This project proposes a freight marketplace where **both sides are represented by agents**, operating under designed market rules rather than free-form haggling. It adds two contributions beyond matching: (1) **coalition formation**, letting small truckers pool trucks to win large contracts, with earnings split fairly using the Shapley value; and (2) a **robustness layer** that tests and defends the market against lying, collusion and prompt injection between agents. The system is evaluated in a simulation built on a real city road network, comparing empty kilometres, waiting time, trucker earnings and fairness against a no-marketplace baseline.

2. Problem Statement

PS: Small truck operators in India lose income and burn fuel on empty return trips or long waits for return loads, because finding, pricing and committing to a return load depends on slow, manual, broker-mediated negotiation that a driver cannot do while driving. Existing digital platforms match loads but leave negotiation to humans or to agents that represent only one side, and small operators remain locked out of large contracts that no single one of them can serve. **How can autonomous agents negotiate return loads and form trucker coalitions on behalf of both sides, fairly, feasibly and safely, with the human retaining final control?**

3. The Problem in Detail

3.1 Why trucks run empty

A truck carries goods from A to B. For the trip back, the driver needs a load that starts near B, goes towards A, fits his spare capacity, matches his timing, and pays enough to justify any detour. Finding it is a search and negotiation problem with incomplete information, done today by phone. The driver faces a bad choice:

- **Return empty:** fuel, tolls, driver wages and vehicle wear are paid with zero revenue. This is common for shorter, city-region trips.
- **Wait for a load:** common for long-haul drivers who cannot afford an empty run; the truck sits idle for hours or days, and round-trip time and income suffer.
- **Use a broker:** faster, but the broker takes a commission and knows market rates better than the driver does.

The cost does not stay with the trucker. Carriers price expected empty running into every quote, so shippers and finally consumers pay for it. Trucking platforms widely cite India's logistics costs at about 13–14% of GDP versus 8–9% in developed economies, with empty running and slow return-load discovery named among the main reasons.

3.2 Example 1: the owner-driver

Raju owns two trucks in Hosur. He delivers 10 tonnes to a Bommasandra warehouse at 1 pm. While driving there, he could not call anyone. At the warehouse, he spends an hour on calls; a broker offers a load at a low rate minus commission. Raju declines and drives 40 km back empty: roughly 12 litres of diesel and half a day's earnings lost. Meanwhile, 9 km away, a Peenya-bound load of 8 tonnes was waiting for a truck.

3.3 Example 2: the small shipper

Meena is the dispatch in-charge at a fabrication unit in Peenya. Every day she calls four or five transporters to move small consignments, negotiating each by phone. She pays full round-trip rates because she has no way to find trucks that are already heading her way empty, trucks that would gladly accept a lower price.

3.4 Example 3: the contract no one can win

A cement plant needs 30 truckloads a week for three months. The contract goes to a large fleet because no single small operator can guarantee that volume. Six operators with five trucks each could serve it together, but coordinating strangers, agreeing on schedules and splitting earnings fairly is practically impossible over the phone. Small operators, the overwhelming majority of the supply, are structurally locked out of the most stable income.

3.5 Who is affected

Stakeholder	Pain today	What they need
Truck owner / driver	Empty runs, idle waiting, broker commissions, cannot negotiate while driving	Profitable return loads found and priced for him, one-tap decision
Shipper (MSME)	Hours of phone calls per load, full round-trip pricing	Fast booking, cheaper return-trip capacity
Platform operator	Matching at scale, fraud, disputes, trust	Fair, auditable, manipulation-resistant market
Economy and environment	Higher logistics cost, wasted diesel, emissions	Better asset utilisation (SDG 9.4)

4. Existing Solutions and the Gap

Load matching is already a commercial product. BlackBuck, a listed Bengaluru company, reports close to a million truck operators on its platform and a loads marketplace inside its app; its founder has stated that 95% of load acceptance comes from small fleet owners with two or three trucks. Many startups and hackathon projects also build "Uber for trucks" matching apps. **A matching app alone is therefore not novel**, and this project does not claim it is.

In the United States, AI voice agents (for example, HappyRobot) now negotiate freight rates at many of the largest brokers. These agents represent the **broker only**; the carrier on the other end is a human with no agent. On the research side, Microsoft Research's Magentic Marketplace (2025) simulated two-sided agent markets and found that frontier models approach optimal welfare only under ideal conditions, degrade sharply with scale, show strong first-proposal bias, and can be manipulated; the authors concluded that human oversight remains critical. Operations research has long studied carrier coalitions with Shapley-value profit sharing (e.g. Krajewska et al., 2008), but assumes a central planner with

full information rather than self-interested agents with private costs.

What exists	Examples	Limitation / how this project differs
Load-matching apps	BlackBuck, other Indian startups, hackathon apps	Humans still search and negotiate; used here only as the base layer
AI agents negotiating freight	HappyRobot and similar (US)	One-sided: broker has an agent, trucker does not. The proposed system represents both sides
Agent marketplace research	Microsoft Magentic Marketplace	Generic domain, no physical constraints. The proposed system adds routing feasibility and designed mechanisms
Carrier coalitions + Shapley	OR literature (Krajewska et al. 2008 and later)	Central planner, full information. The proposed system forms coalitions via negotiating agents with private information
Agent security	Emerging research	Rarely tested in a concrete economic market. The proposed system measures attacks and defences quantitatively

Lesson from Convoy: the US digital freight broker Convoy raised over a billion dollars and shut down in 2023 amid a freight recession. Technology alone does not guarantee adoption; trust, payment security and market economics matter. This project therefore builds human approval, reputation and payment guarantees into the design rather than treating them as afterthoughts.

5. Proposed Solution (Tentative)

The proposed system is an agent-mediated freight marketplace. The mental model is a ride-hailing app for trucks, with one key difference: in Rapido or Uber the platform sets the fare and assigns the nearest request; in this system **there is no fixed fare**. Each price emerges from negotiation between the trucker's agent and the shipper's agent under market rules that make honest behaviour the best strategy. By the time the driver's phone buzzes, the bargaining is already done.

Five design principles:

- **Every party has its own agent.** Each agent guards its owner's private limits (minimum price, deadlines, personal constraints) and never reveals them directly.
- **LLMs negotiate; code decides what is true.** Feasibility (time windows, capacity, detours) is checked by a deterministic optimiser with veto power. Fair splits are computed mathematically, not argued.
- **Rules, not free-for-all.** Structured auctions and protocols replace open-ended haggling, directly addressing the failure modes found in agent-market research.
- **The human decides.** Owners set limits once; agents search and bargain; nothing becomes binding without a one-tap yes.
- **Trust is designed in.** Reputation scores and escrow-style payment guarantees address the trucker's biggest fear: not getting paid.

6. How the Solution Works

6.1 The agents

Component	Type	Role
Trucker agent (one per truck)	LLM agent	Knows location, spare capacity, costs and the owner's rules; evaluates loads, bids, negotiates, joins coalitions
Shipper agent (one per shipper)	LLM agent	Turns messy requests (WhatsApp text, voice) into structured loads; runs the reverse auction; negotiates terms
Marketplace (broker) agent	LLM agent, single	Shortlists candidate matches, proposes coalitions for large contracts, re-opens deals after disruptions
Route optimiser	Deterministic (OR-Tools)	Checks every deal for time windows, capacity and detour cost; vetoes infeasible deals
Mechanism and reputation module	Deterministic code	Runs auctions, computes Shapley splits, scores reliability, flags suspicious bidding

6.2 Life of one deal

Step	What happens
1. Request	Meena's clerk sends: "8 ton castings Bommasandra to Peenya, today before 6, around 10 rupees per km." The shipper agent converts this into a structured load.
2. Shortlist	The marketplace agent finds five trucks returning in roughly that direction with enough spare capacity.
3. Evaluate	Raju's agent calls the route tool (9 km detour, arrival 5:20 pm), prices the extra diesel, and checks Raju's rules (minimum ₹12/km, home by Sunday). Worth it, so it bids.
4. Negotiate	Bids go into a reverse auction; the shipper agent counters; the deal settles at ₹12.5/km within a few rounds.
5. Verify	The optimiser confirms feasibility; the reputation module checks both parties.
6. Approve	Raju receives the finished deal with reasoning and taps Accept.
7. Adapt	If the truck breaks down, the agent reports it and the shipper agent re-opens the auction with the next-best truck, with no phone calls.

6.3 What the driver sees

Load: Bommasandra → Peenya, 8 tonnes

₹12.5/km (agent opened at ₹14, shipper countered ₹11)

Pickup 3:00 pm · 9 km detour · arrive 5:20 pm

Why recommended: same direction as your return trip; about ₹2,900 extra for roughly 40 minutes more driving; shipper rating 4.6, pays on time.

Also considered: one offer at ₹10/km (below your minimum); one that would miss your Sunday return.

[Accept] [Reject] [Keep negotiating] · offer holds for 10 minutes

Notifications are filtered hard: the driver sees only the best one or two finished deals, each with its reasoning and the rejected alternatives, so he can trust the agent without redoing its work.

6.4 Trucker coalitions

For large contracts, the marketplace agent proposes groups of trucks whose combined capacity and schedules can serve the contract. Trucker agents negotiate whether to join, the

optimiser builds the joint schedule, and earnings are split using the Shapley value, which pays each member according to their average marginal contribution across all possible groups. The driver receives: "Join a 6-truck group for the weekly cement contract. Your share: ₹18,400 for 3 trips this week. Split calculated by contribution." This is where agents are not merely convenient: coordinating and fairly paying a group of strangers is practically impossible manually.

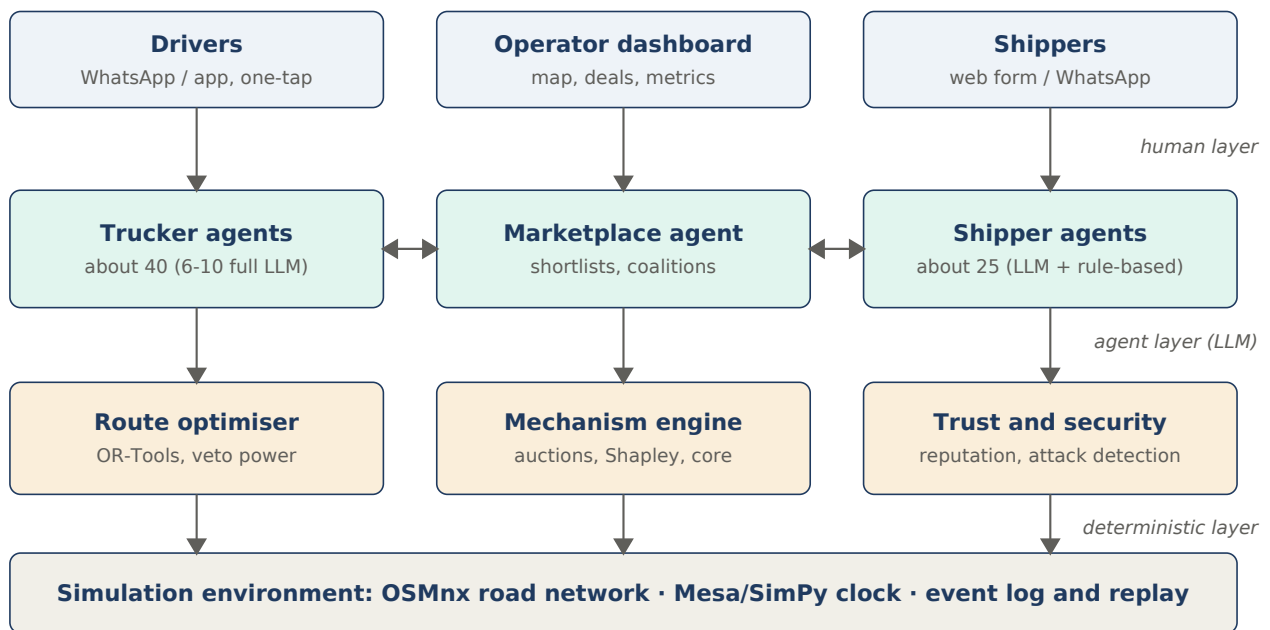
6.5 Why agents and not just an algorithm?

- **Private information:** operators will not hand their true minimum price to a central system; agents let each side negotiate while keeping it private.
- **Messy inputs:** loads and preferences arrive as WhatsApp messages and voice notes in Kannada, Hindi or English.
- **Fuzzy preferences:** "home by Sunday for a family function" is easy for an LLM to reason with, hard to encode as a formula.
- **Change:** breakdowns, delays and new loads demand continuous re-planning.

The optimiser handles what can be calculated exactly; agents handle what needs judgement, communication and negotiation. That split is the core idea of the project.

7. Technical Details

7.1 Architecture



Agents exchange structured JSON messages (OFFER, COUNTER, ACCEPT, REJECT, JOIN_COALITION) over a defined protocol, capped at a few rounds per deal. Free-text reasoning is kept inside each agent; only typed fields cross between agents, which is also the first line of defence against prompt injection.

7.2 Game theory and market design

Situation	Theory used	What we implement / measure
One trucker vs one shipper	Bargaining theory (reservation prices, outside options, Nash bargaining solution)	Benchmark: do LLM agents reach fair splits of the surplus, or does one side get exploited?

Situation	Theory used	What we implement / measure
Many trucks, one load	Reverse auctions: first-price vs second-price (Vickrey)	Vickrey makes bidding true cost the dominant strategy; compare honesty and prices
Many trucks, many loads (bundles)	Combinatorial auctions, VCG mechanism	Winner determination is NP-hard: solved with integer programming in OR-Tools
Coalitions for large contracts	Cooperative game theory: characteristic function, Shapley value, the core	Monte Carlo Shapley approximation; test stability (does any subgroup prefer to break away?)
Collusion	Bid-rigging patterns from auction theory	Detection features on bid histories

A key design discussion: the Myerson-Satterthwaite theorem shows that no two-sided trading mechanism can be simultaneously efficient, truthful, voluntary and budget-balanced. The system must choose which property to relax, justify the choice, and show its consequences experimentally.

7.3 Optimisation

Feasibility checks and scheduling are modelled as a pickup-and-delivery problem with time windows and capacity constraints, solved with Google OR-Tools. The optimiser computes detour distance, arrival times and marginal cost for each proposed deal, and builds joint schedules for coalitions.

7.4 Robustness and security experiments

- **Attacks:** cost misreporting, collusive bidding among trucker agents, prompt injection hidden in offer text, and fear/urgency manipulation.
- **Defences:** typed message schema with sanitisation, truthful auction rules, reputation-weighted matching, and anomaly detection on bidding patterns.
- **Metric:** attack success rate and welfare loss, before and after each defence.

7.5 Simulation setup

- Real road network of an Indian industrial region (e.g. Peenya, Bommasandra, Hosur, Whitefield) from OpenStreetMap via OSMnx.
- ~40 trucks and ~25 shippers in the live demo; only 6–10 are full LLM agents, the rest use rule-based strategies (e.g. accept above cost + 15%). This controls API cost and yields a built-in comparison of LLM vs rule-based negotiators.
- Offline scaling runs with a few hundred rule-based agents to test behaviour at scale.
- Fixed random seeds so the baseline and marketplace runs face identical trucks, orders and timings.

7.6 Evaluation metrics

Category	Metrics
Efficiency	Empty km, truck waiting time, fuel and CO ₂ saved, share of return trips loaded
Economics	Trucker earnings, shipper cost, total welfare vs optimal (computed by the optimiser with full information)
Fairness	Surplus split vs Nash bargaining benchmark; Shapley shares; coalition stability (core)
Robustness	Attack success rate, welfare loss under attack, detection precision/recall
Agent quality	First-offer bias, rounds to agreement, LLM vs rule-based negotiator outcomes

7.7 Technology stack

Layer	Tools
Simulation	Python, Mesa or SimPy, OSMnx, NetworkX
Agents	LLM API with tool calling; structured JSON protocol; optionally reuse components of the open-source Magentic Marketplace
Optimisation and mechanisms	Google OR-Tools (routing + integer programming), NumPy for Shapley sampling
Backend	FastAPI, WebSocket event streaming, PostgreSQL event log
Frontend	React with Leaflet or deck.gl map; mobile-style notification view for drivers

7.8 Demo plan

- **Act 1, baseline:** a simulated day with no marketplace; empty return routes shown in red, counters for empty km, fuel and lost income.
- **Act 2, marketplace on:** same seed; agents negotiate; return trips turn green; live negotiation feed.
- **Act 3, one deal up close:** offer, counter, optimiser verdict, and the driver's notification with reasoning.
- **Act 4, stress tests:** a lying agent, a colluding group, an injection attempt and a breakdown, followed by aggregate results over 20–30 simulated days.
- A recorded replay mode protects the live evaluation against API or network failure.

8. Expected Outcomes and Limitations

Expected outcomes: a measurable reduction in empty kilometres and waiting time against the baseline; evidence on whether structured mechanisms outperform free-form LLM negotiation on fairness and honesty; a working coalition mechanism with fair, stable splits; and quantified attack and defence results for agent-to-agent markets.

Limitations: results come from simulation with synthetic demand, so absolute savings will not transfer directly to the real market; adoption depends on trust, payment guarantees and platform economics that a semester project cannot fully test; and LLM behaviour varies between models and versions, so results should be reported across more than one model.

9. References

- [1] Fleetx, "Backhaul Optimization as a Profitability Booster" (industry estimate of empty running in India).
- [2] IRADE, "Fuel Subsidy and Transport Policy Brief" (empty returns vs waiting behaviour of Indian truckers).
- [3] TechCrunch and YourStory coverage of BlackBuck (Zinka Logistics): platform scale and small-fleet load acceptance.
- [4] CNBC / Supply Chain Dive (2023), coverage of Convoy's shutdown.
- [5] HappyRobot case studies and funding coverage (2024–2026): AI agents in US freight brokerage.
- [6] Bansal et al., "Magentic Marketplace: An Open-Source Environment for Studying Agentic Markets", Microsoft Research / ASU, arXiv:2510.25779 (2025).
- [7] Krajewska, Kopfer, Laporte, Ropke, Zaccour, "Horizontal cooperation among freight carriers: request allocation and profit sharing", J. Operational Research Society 59 (2008).
- [8] Roughgarden, T., Twenty Lectures on Algorithmic Game Theory, Cambridge University Press (2016).

- [9] Nisan, Roughgarden, Tardos, Vazirani (eds.), *Algorithmic Game Theory*, Cambridge University Press (2007).
- [10] Myerson, R. and Satterthwaite, M., "Efficient mechanisms for bilateral trading", *Journal of Economic Theory* 29 (1983).